CRA Tax Support Assistant for UofT Students

PROJECT PROPOSAL · APPLICATIONS OF LARGE LANGUAGE MODELS · WINTER 2026

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UNIVERSITY OF TORONTO STUDENTS' UNION

1 Knowledge Base Selection

PRIMARY: CRA (CANADA.CA)

P105 Students & Income Tax, GST/HST credit, T2202, Schedule 11, international student tax guide, CRA PDF publications (T4044, RC4064)

≈ 60–80 HTML pages + 10–15 PDF guides (100+ pages each)

SUPPLEMENTARY: UOFT RESOURCES

UTSU Tax Clinic handouts, Centre for International Experience guides, UofT Registrar tax pages, CVITP training materials

≈ 15–20 web pages + PDF worksheets

100+	✓	HTML+PDF
TOTAL SOURCES	PUBLIC ACCESS	FORMATS

Backup Sources

- **Wealthsimple Tax** student guide, step-by-step filing walkthrough
- **TurboTax** student tax tips, deduction checklist
- **Woodsworth Tax Clinic** UofT-specific handouts & FAQ

Why This Domain?

- International students at UofT pay among Canada's highest tuition yet miss credits worth **thousands**: tuition carry-forward, GST/HST credit, Ontario Trillium Benefit.
- Team includes a former **UTSU Tax Clinic CVITP volunteer**, who has seen firsthand: students arriving unprepared, missing documents, unaware of eligibility.
- UTSU serves **400+ students/yr** with ~20 volunteers, Mar–Apr only. Our agent offers **24/7** pre-screening & document prep, making every volunteer hour count.
- *Natural guardrail:* agent is **not** a licensed tax advisor, always disclaims & redirects.

Risks & Mitigation

- **CRA page restructure:** Archived snapshots via Wayback Machine; PDF publications are stable and version-controlled.
- **Tax advice liability:** Every response includes disclaimer; agent refuses to "file" or give definitive advice.
- **Model hallucination:** Strict RAG grounding; refuse if retrieval confidence < threshold; cite source URL.

2 Planned Agent Capabilities

Tax Estimate — Single-turn
`calculate_tax_estimate`
Income + province → federal & provincial bracket estimate
"I made $18k as a TA, how much do I owe?"

Benefit Eligibility — 4 turns
`check_benefit_eligibility`
GST/HST Credit, OTB, CCB, tuition carry-forward
"Am I eligible for the GST/HST credit?"

Filing Reminder — 2 turns
`create_filing_reminder`
Mock reminder email for Apr 30 or UTSU booking
"Remind me before the Apr 30 deadline"

Book UTSU Appt — 5–6 turns
`book_tax_clinic_appt`
CVITP eligibility screen + doc prep + Calendly link
"Book me a tax clinic appointment"

book_tax_clinic_appt — example flow

U: I want to book a tax appointment at UTSU
Bot: Sure! Are you undergrad or grad?
U: Grad, income ~$25k
Bot: Self-employed / capital gains / foreign income >$1k?
U: No
Bot: ✅ Eligible! Do you have SIN, T2202, T4 ready?
U: SIN + T4, no T2202
✅ Checklist generated → Calendly booking link

Vision

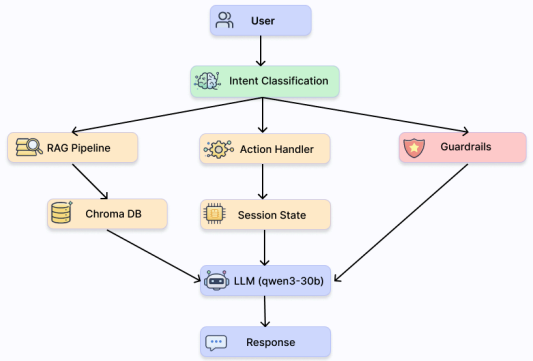
utsu.ca/tax · Analytics · 24/7 service · 中文 · 한국어 · UTSU booking · हिन्दी · auto-fill · multilingual · CRA API · volunteer dashboard · Français · pre-screening · GST/HST · walk-in → widget · 3x capacity

Guardrails

- **Reject:** "help me avoid taxes," "fill out my return"
- **Disclaim:** every estimate → "general info only"
- **Redirect:** "Am I audited?" → CRA 1-800-959-8281
- **Scope:** Canadian student tax topics only

3 Technical Approach

CRA Tax Support Assistant



Pipeline

Intent (Classify & route) → RAG (Retrieve & rank) → Action (Execute & persist) → LLM (Cite & disclaim)

qwen3-30b-a3b-fp8 · LangChain · LangGraph · ChromaDB · sentence-transformers · SQLite · Streamlit · BeautifulSoup · Railway (bonus)

Retrieval Accuracy	Answer Faithfulness	Guardrail Compliance	Tool Routing	Latency (e2e)
≥ 85%	≥ 90%	≥ 95%	≥ 80%	< 4s

MAR 14–20	MAR 21–27	MAR 28–APR 3	APR 4–10	APR 13–17
Scrape & chunk CRA + UofT pages	Embed, index & RAG pipeline	Agent logic, tools & multi-turn flows	UI, guardrails eval & deploy	Presentations & TA demos